Structure Factor report

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In this report are the main results of the structure factor analysis.

Characteristic distances in the system. The distance range for the pairwise interaction is $r_b \in [0.4, 0.6]$. The total distance between the center of two central particles that aire in a pairwise interaction is $r_{\text{int}} = 0.8 + r_b \therefore r_{\text{int}} \in [1.2, 1.4]$. By doind some math we get that if we increase the amount of central particles in a line within the pairwise distance interaction range, the distance increases as follows, $d_{\text{int}}(N_{\text{central}}) = (N_{\text{central}} - 1)\,(2\sigma_p + r_b)$.

Domains for the fits. From the characteristic distances we can define the following spatial domains $\lambda_1 \in \left[7/2\sigma_p, 21/2\sigma_p\right]$ and $\lambda_2 \in \left[21/2\sigma_p, L/4\right]$.

About the fit. The experimental data was subjected to a fit

$$f(|\vec{q}|) = \frac{p_1}{|\vec{q}|^{p_2}}.$$

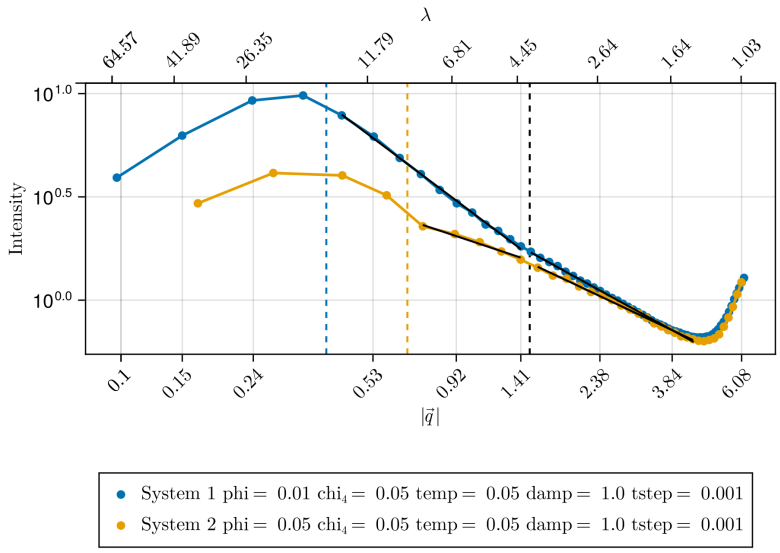


Figure 1: Structure factor of two different systems at the same experiment.